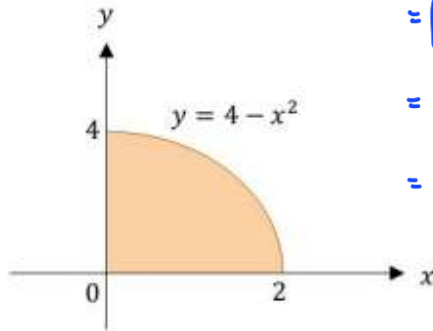


TUTORIAL 5: APPLICATIONS OF INTEGRATION

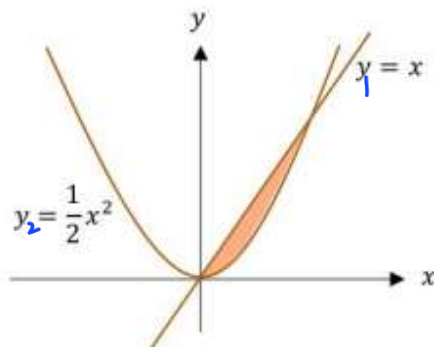
1. Find the area of shaded region.

a.



$$\begin{aligned} \text{a) } A &= \int_0^2 y \, dx \\ &= \int_0^2 (4 - x^2) \, dx \\ &= \left[4x - \frac{x^3}{3} \right]_0^2 \\ &= \left(8 - \frac{8}{3} \right) - (0) \\ &= \frac{16}{3} \text{ unit}^2 \end{aligned}$$

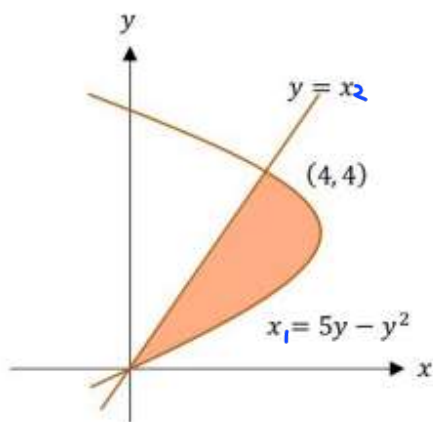
b.



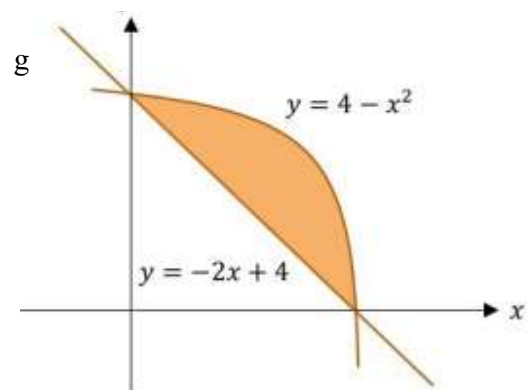
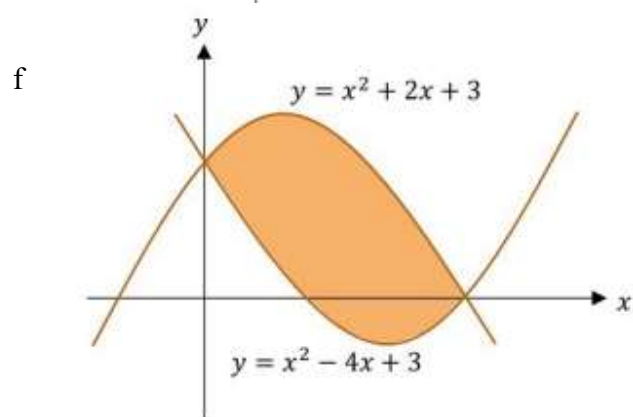
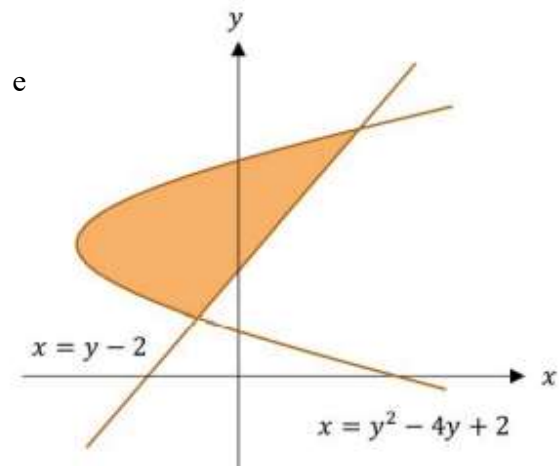
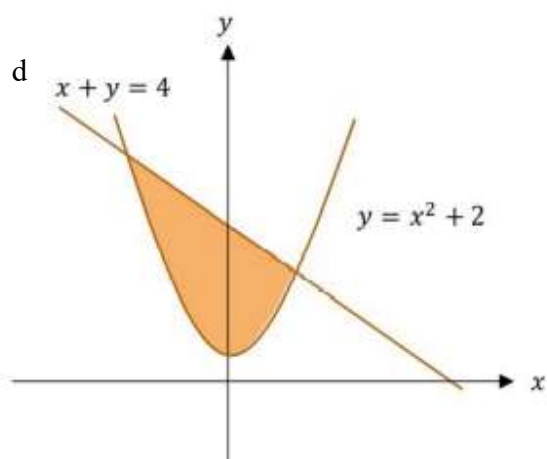
$$\begin{aligned} \text{(b) } y_1 &= y_2, \quad x = \frac{1}{2}x^2 \\ \frac{1}{2}x^2 - x &= 0 \\ x &= 0, \quad x = 2 \end{aligned}$$

$$\begin{aligned} A &= \int_0^2 (y_1 - y_2) \, dx \\ &= \int_0^2 \left(x - \frac{1}{2}x^2 \right) \, dx \\ &= \left[\frac{x^2}{2} - \frac{x^3}{6} \right]_0^2 \\ &= \left(2 - \frac{8}{6} \right) - \left(\frac{0}{2} - \frac{0}{6} \right) \\ &= 0.67 \text{ unit}^2 \end{aligned}$$

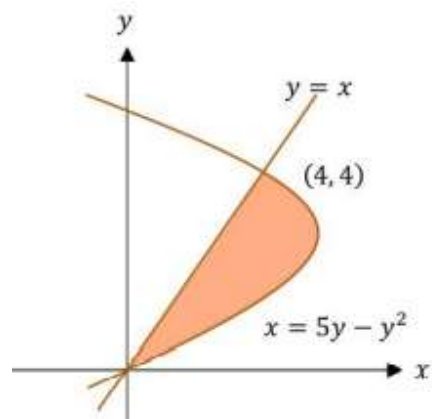
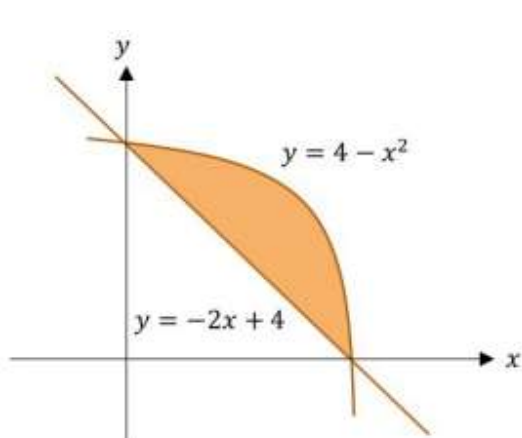
c.



$$\begin{aligned} \text{(c) } y &= 0, \quad y = 4 \\ A &= \int_0^4 (x_1 - x_2) \, dy \\ &= \int_0^4 (5y - y^2 - y) \, dy \\ &= \left[2y^2 - \frac{y^3}{3} \right]_0^4 \\ &= \left[32 - \frac{64}{3} \right] - [0 - 0] \\ &= \frac{32}{3} \text{ unit}^2 \end{aligned}$$



2. Find the volume of the shaded region.



$$\begin{aligned}
 (d) \quad y_1 &= y_2 & A &= \int_{-2}^1 4 - x - x^2 - 2 \, dx \\
 4 - x &= x^2 + 2 & &= \int_{-2}^1 -x^2 - x + 2 \, dx \\
 x^2 + x - 2 &= 0 & &= \left[-\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_{-2}^1 \\
 x &= -2, x = 1 & &= \left(-\frac{1}{3} - \frac{1}{2} + 2 \right) - \left(-\frac{8}{3} - \frac{4}{2} - 4 \right) \\
 & & &= \frac{7}{6} + \frac{10}{3} \\
 & & &= \frac{9}{2} \text{ unit}^2
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad y - 2 &= y^2 - 4y + 2 & A &= \int_1^4 y - 2 - y^2 + 4y - 2 \, dy \\
 y^2 - 5y + 4 &= 0 & &= \int_1^4 -y^2 + 5y - 4 \, dy \\
 y &= 1, y = 4 & &= \left[-\frac{y^3}{3} + \frac{5y^2}{2} - 4y \right]_1^4 \\
 & & &= \left[-\frac{64}{3} + \frac{80}{2} - 16 \right] - \left[-\frac{1}{3} + \frac{5}{2} - 4 \right] \\
 & & &= \frac{8}{3} + \frac{11}{6} \\
 & & &= \frac{9}{2} \text{ unit}^2
 \end{aligned}$$

$$\begin{aligned}
 (f) \quad y &= 0, 0 = x^2 - 4x + 3 & A &= \int_0^3 y_1 - y_2 \, dx \\
 x &= 3 & &= \int_0^3 x^2 + 2x + 3 - x^2 + 4x - 3 \, dx \\
 & & &= \int_0^3 6x \, dx \\
 & & &= [3x^2]_0^3 \\
 & & &= 27 \text{ unit}^2
 \end{aligned}$$

$$\begin{aligned}
 (g) \quad 4 - x^2 &= -2x + 4 & A &= \int_0^2 4 - x^2 + 2x - 4 \, dx \\
 x^2 - 2x &= 0 & &= \left[-\frac{x^3}{3} + x^2 \right]_0^2 \\
 x &= 0, x = 2 & &= \left(-\frac{8}{3} + 4 \right) - (0 - 0) \\
 & & &= \frac{4}{3} \text{ unit}^2
 \end{aligned}$$

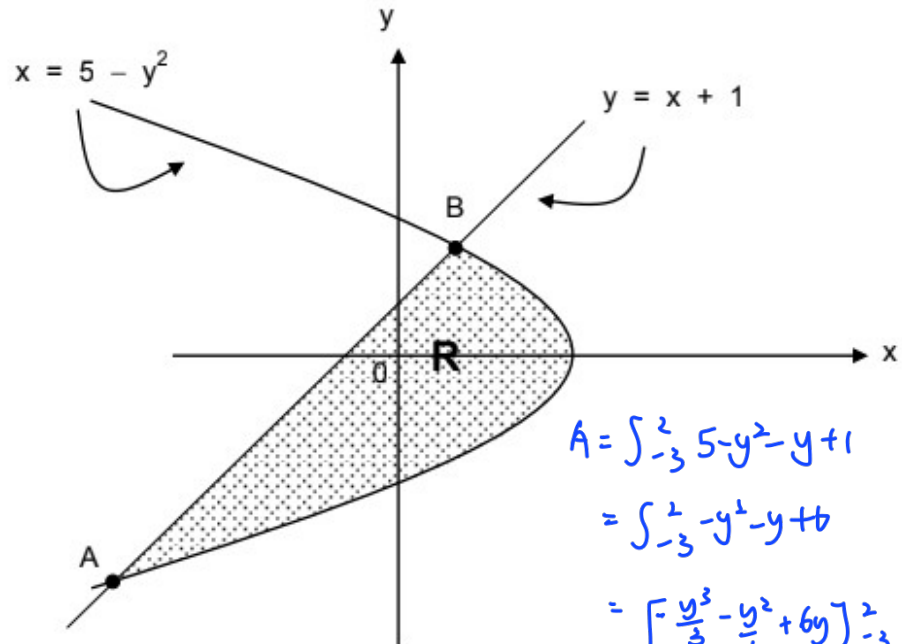
$$\begin{aligned}
 2) (a) \quad V &= \pi \int_0^2 (4 - x^2)^2 - (-2x + 4)^2 \, dx \\
 &= \pi \int_0^2 16 - 8x^2 + x^4 - 4x^2 + 16x - 16 \, dx \\
 &= \pi \int_0^2 x^4 - 12x^2 + 16x \, dx \\
 &= \pi \left[\frac{x^5}{5} - 4x^3 + 8x^2 \right]_0^2 \\
 &= \frac{32}{5} \pi \text{ unit}^3
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad V &= \pi \int_0^4 (5y - y^2)^2 - (y)^2 \, dy \\
 &= \pi \int_0^4 25y^2 - 10y^3 + y^4 - y^2 \, dy \\
 &= \pi \int_0^4 y^4 - 10y^3 + 24y^2 \, dy \\
 &= \frac{384}{5} \pi \text{ unit}^3
 \end{aligned}$$

The region R is bounded by the curves $x = 5 - y^2$ and $y = x + 1$.

$$\begin{aligned}x_1 &= x_2 \\5 - y^2 &= y - 1 \\y^2 + y - 6 &= 0 \\y &= 2, y = -3 \\x &= 1, x = -4\end{aligned}$$

$$\begin{aligned}A &(-4, -3) \\B &(1, 2)\end{aligned}$$

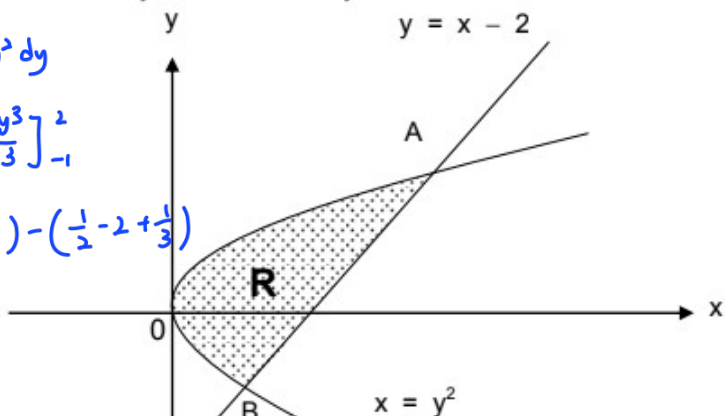


$$\begin{aligned}A &= \int_{-3}^2 (5 - y^2 - y + 1) dy \\&= \int_{-3}^2 (-y^2 - y + 6) dy \\&= \left[-\frac{y^3}{3} - \frac{y^2}{2} + 6y \right]_{-3}^2 \\&= \frac{125}{6} \text{ unit}^2\end{aligned}$$

- Find the coordinates of A and B.
- Determine the area of the shaded region R.

The diagram shows the region R bounded by the curve $x = y^2$, which meet at the points A and B.

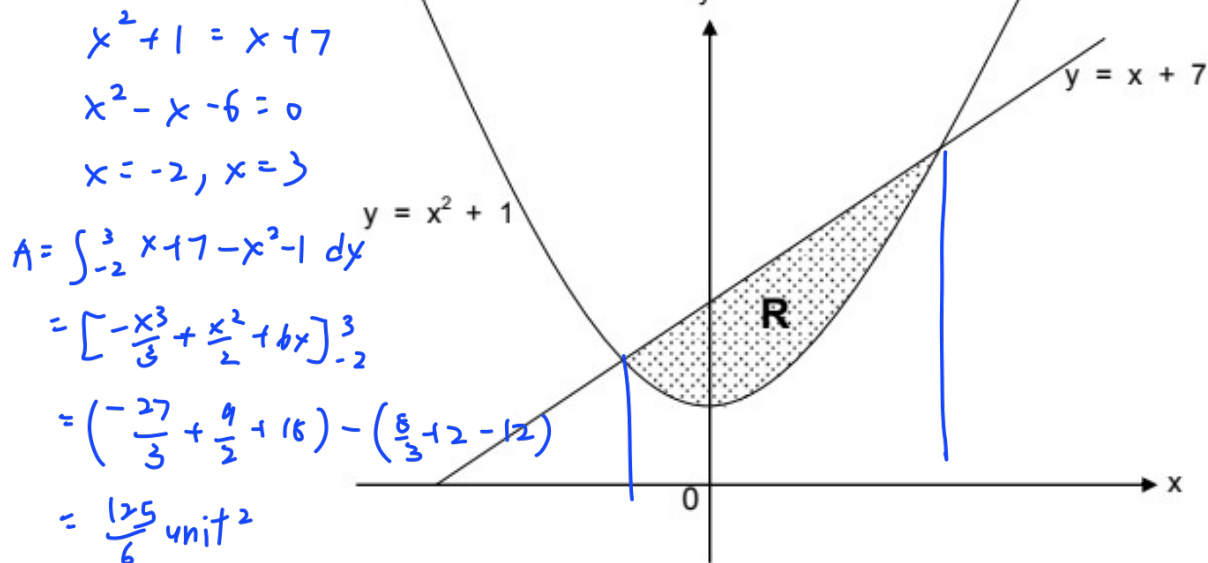
$$\begin{aligned}x_1 &= y + 2 & A &= \int_{-1}^2 (y + 2 - y^2) dy \\x_2 &= y^2 & &= \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2 \\y^2 - y - 2 &= 0 & &= \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) \\y &= -1, y = 2 & &= \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) \\x &= 1, x = 4 & &= \frac{9}{2} \text{ unit}^2 \\A &(4, 2), B(1, -1) & &\end{aligned}$$



- Find the coordinates of A and B.
- Determine the area of the shaded region.
- Find the volume generated when R is rotated through 360° about the y-axis.

$$\begin{aligned}V &= \pi \int_{-1}^2 (y + 2)^2 - (y^2)^2 dy \\&= \pi \int_{-1}^2 (y^2 + 4y + 4) - y^4 dy \\&= \left[-\frac{y^5}{5} + \frac{y^3}{3} + 2y^2 + 4y \right]_{-1}^2 \\&= \frac{72}{5} \pi \text{ unit}^3\end{aligned}$$

The diagram below shows the region R bounded by two functions $y = x^2 + 1$ and $y = x + 7$.



- Find the area of the region R.
- Find the volume of the solid obtained by revolving the region R about x-axis

$$V = \pi \int_{-2}^3 (x + 7)^2 - (x^2 + 1)^2 dx$$

$$= \pi \int_{-2}^3 (x^2 + 14x + 49 - x^4 - 2x^2 - 1) dx$$

$$= \left[-\frac{x^5}{5} - \frac{x^3}{3} + 7x^2 + 48x \right]_{-2}^3$$

$$= \frac{625}{5} \pi \text{ unit}^3$$

Handwritten blue note next to the integral:

$$-x^4 - x^2 + 14x + 48$$